MILITARY
SPECIFICATION

MIL-B-11453A(SigC)
Amendment 4
7 March 1961

BATTERY, STORAGE BB-401( )/U

1. Paragraph 2.1: Under drawings replace the nomenclature "BB-401/U" with "BB-401( )/U."

2. Paragraph 3.2.2 Vents, line 2: Replace "2 ± 1.0 psi" with "2 ± 1.5 psi."

3. Paragraph 3.9 Electrical requirements.- Add the following sub-paragraph:

3.9.1 Accelerated electrical test requirements.- At the end of 15 seconds of discharge through the 2.8 ohm resistor, the loaded voltage shall be no less than 28.5 volts. The time tolerance is plus or minus 1 second, however, if the voltage is still above 28.5 when time exceeds the 16 seconds, the sample will be considered acceptable. If time of voltage measurement is less than 14 seconds, or above 16 seconds and the voltage below 28.5 volts, the test will be rerun. This requirement will only apply to the full charge capacity discharge test and the short term (30 day) stand-by capacity test.

4. Paragraph 3.11 Stand-by capacity.- Replace with the following:

3.11 Stand-by capacity.- The equipment shall be capable of meeting the requirements of 3.9 and 3.9.1 after being charged with a trickle charge of 1.4 volts per cell in an average temperature range of from + 70°F to + 125°F for the following time periods:

Short term period - 30 days (see 4.12.1)
Long term period - 1 year (see 4.12.2)

5. Paragraph 3.15.1 General.- Marking shall be in accordance with the requirements of Group I of Specification MIL-M-13231. In addition, marking on plates specified in 3.15.2 shall be resistant to and insoluble in Potassium Hydroxide.

6. Page 5. Add the following between 4 and 4.1:

Contractor's responsibility.- The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

FSC 6140

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ACTION TAKEN

7. Paragraph 4.3.1 Group A inspection.- Modify Table II as follows:

(a) In requirement paragraph column, add "and 3.9.1" after 3.9.

(b) In inspection paragraph column, add "and 4.7.1" after 4.7.

(c) Add the following:

Inspection	Req Par	Insp Par	AQL (%) Major	AQL (%) Minor
Vent Plug	3.2.2	4.5	0.65	

8. Paragraph 4.3.3 Group C inspection.- Add the following to table III:

Inspection	Req Para	Insp Para
Stand-by capacity (Short term)	3.11	4.12.1

9. Paragraph 4.3.3.1 Sampling for inspection of equipment: Add the following sub-paragraph:

4.3.3.1.1 Sampling for inspection of cells (BB-403( )/U).- Forty eight cells (BB-403( )/U) shall be selected without regard to their quality from each lot released for shipment.

10. Paragraph 4.4 Government Laboratory Inspection.- Replace with the following:

4.4 Government Laboratory Inspection.- This inspection will be performed at a Government Laboratory and shall consist of the preproduction inspection, group A, group C and Government laboratory inspection specified in Tables I, II, III and IV, respectively.

11. Table IV Laboratory Inspection.- Replace Stand-by Life Test with the following:

Inspection	Req. Para.	Insp. Para.
Stand-by capacity test (Long term)	3.11	4.12.2

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12. Paragraph 4.4.1 Number of samples.- Replace with the following:

4.4.1 Number of samples.- The total number of samples to be selected from production and furnished to the Government laboratory shall be as follows:

BB-401( )/U		BB-403( )/U	
Total number of batteries on release	Total number of samples to be selected	Total number of batteries on release	Total number of samples to be selected
20-100	2	1-5000	24
101-500	4	5,001-10,000	48
501-1000	6	10,001-20,000	72
1001-2000	8		
2001-4000	10		
Plus 2 additional batteries for each 1000 additional batteries or part thereof over and above 4000 batteries on release.		Plus 24 additional batteries for each additional 10,000 batteries or part thereof over and above 20,000 batteries on release.	

13. Paragraph 4.5 Vent plug test, line 4: Replace "1 $^{+0}_{-.1}$ psi" with "1 $^{+0}_{-.5}$ psi". Line 5: Replace "3 $^{+.1}_{-0}$ psi" with "3 $^{+.5}_{-0}$ psi".

14. Paragraph 4.6 Test conditions.- Modify as follows:

(a) Insert 4.7.1 between 4.7 and 4.8.
(b) Add the following:

4.6.1 Foaming.- Any battery which, while being charged in accordance with the instruction plate, generates foam sufficient to issue from the vents in such a manner that foam from two adjacent cells coalesce, or that foam from one cell spills over and touches an adjacent cell, shall be rejected.

15. Paragraph 4.7 Full charge capacity discharge.- Add the following sub-paragraph:

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4.7.1 Accelerated electrical test.- The voltage shall be read and recorded between 14 and 16 seconds after initiation of discharges covered in 4.7 and 4.12.1. The voltage shall meet the requirements of paragraph 3.9.1.

16. Paragraph 4.12 Stand-by capacity test.- Replace with the following:

4.12 Stand-by capacity test.-

4.12.1 Short-term test.- The batteries after being fully charged shall be maintained at 33.6 ± 0.3 volts for 30 days at a temperature of 75 ± 7°F, 50 ± 5% RH with normal maintenance as specified by the instruction plate (3.15.2.2). At the end of the 30 day period, the batteries shall be tested in accordance with 4.7 and 4.7.1.

4.12.2 Long-term test.- The batteries after being fully charged shall be maintained on a trickle charge at 33.6 ± 0.3 volts for one year at a temperature of 75 ± 7°F, 50 ± 5% RH with normal maintenance as specified by the instruction plate (3.15.2.2). At the end of the one year period, the batteries shall be tested in accordance with 4.7.

17. Paragraph 4.13.1 Electrical leakage test for cells.- Add the following to the end of the paragraph:

In the event that the sealing line is level with or above the terminal studs, the battery shall be tested by attaching one probe of the ohm-meter to one of the terminal studs and the other end to a small wad of wet cotton. The probe with the wet cotton shall be slowly moved along all seal lines. Any reading on the ohm-meter shall be cause for rejection of the cell.

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